

FX Market Regime Detection Using Hidden Markov Models

Objective

To identify persistent macro market regimes in the foreign exchange market using rolling cross-currency correlation structures and Hidden Markov Models (HMMs).

Dataset

- 9 Major FX Pairs
- Daily Data
- June 2006 – June 2026
- 20-Day Rolling Correlations
- 36 Unique Currency Relationships

Key Findings

The model identified two statistically distinct market regimes:

Metric	Regime 0	Regime 1
Average Correlation	0.4151	0.6092
Sample Prevalence	52.63%	47.37%
Average Duration	60.1 Days	38.6 Days

The estimated transition probabilities exceeded 97% for both regimes, indicating strong persistence and suggesting that FX market states evolve gradually rather than randomly.

Practical Implication

Changes in cross-market correlation structures can be used as an objective filter for:

- Risk management
- Position sizing
- Strategy selection
- Macro regime classification

Problem & Research Question

Most traders evaluate markets through individual price charts.

However, market behavior changes significantly across different macro environments.

Strategies that perform well during one environment frequently fail during another.

The challenge is determining:

When has the underlying market environment changed?

This study investigates whether macro FX regimes can be identified using only the relationships between currencies themselves.

Research Question

Can rolling cross-currency correlations reveal persistent hidden market states?

If so:

- How often do these states occur?
- How long do they persist?
- Can they be used as an objective regime filter?

Methodology

Data

Currencies Included:

- AUDUSD
- CADUSD
- CHFUSD
- EURUSD
- GBPUSD
- JPYUSD
- NOKUSD
- NZDUSD
- SEKUSD

Processing Pipeline

1. Download daily FX prices.
2. Calculate daily returns.
3. Standardize returns using z-scores.
4. Compute 20-day rolling correlation matrices.
5. Extract 36 unique pairwise correlations.
6. Calculate average market correlation.
7. Apply a two-state Gaussian Hidden Markov Model.

Model Parameters

- Hidden States: 2
- Covariance Type: Full
- Maximum Iterations: 1000
- Random Seed: 42

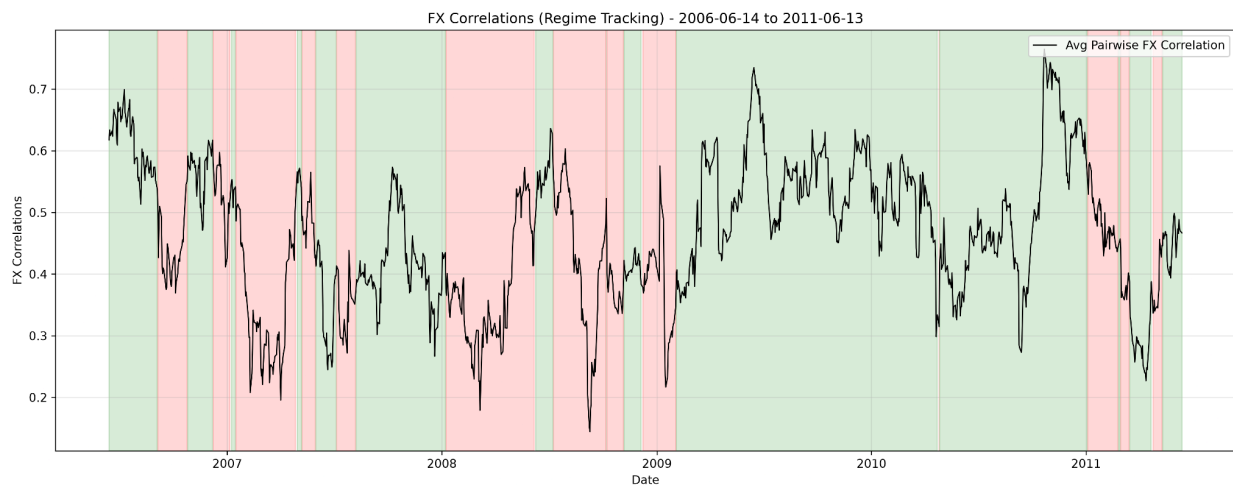
Regime Classification

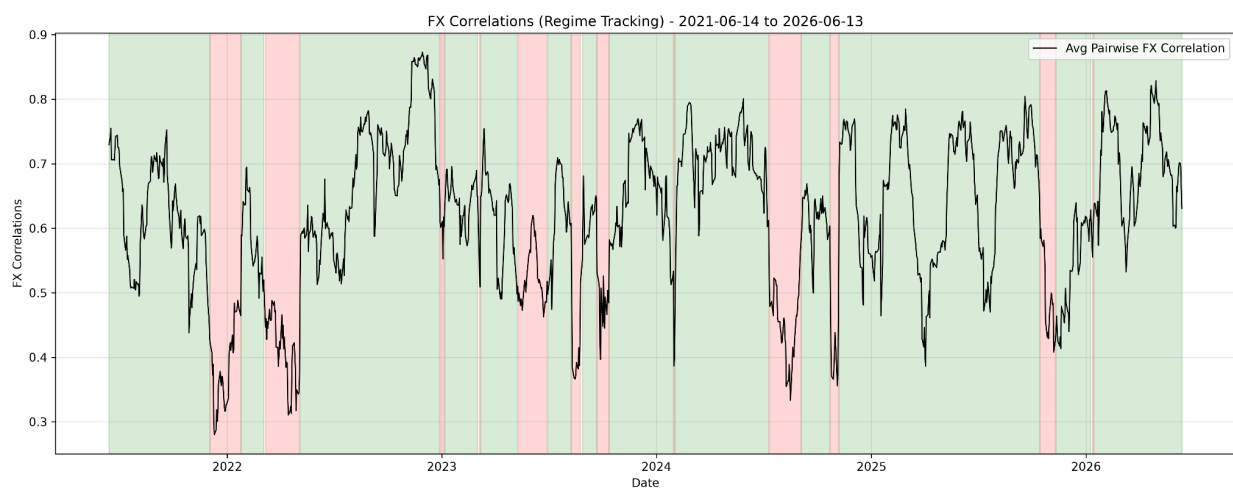
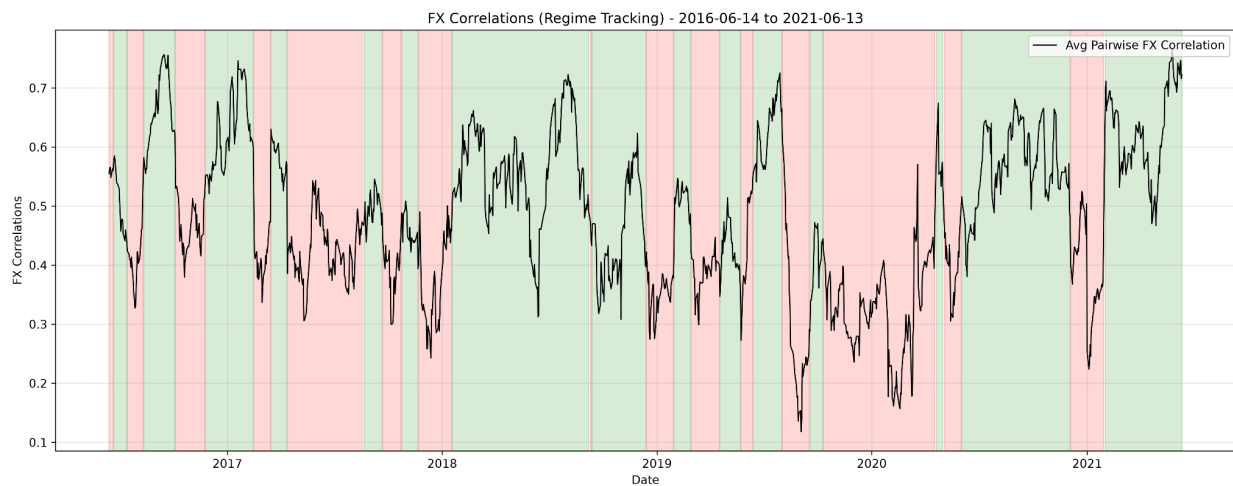
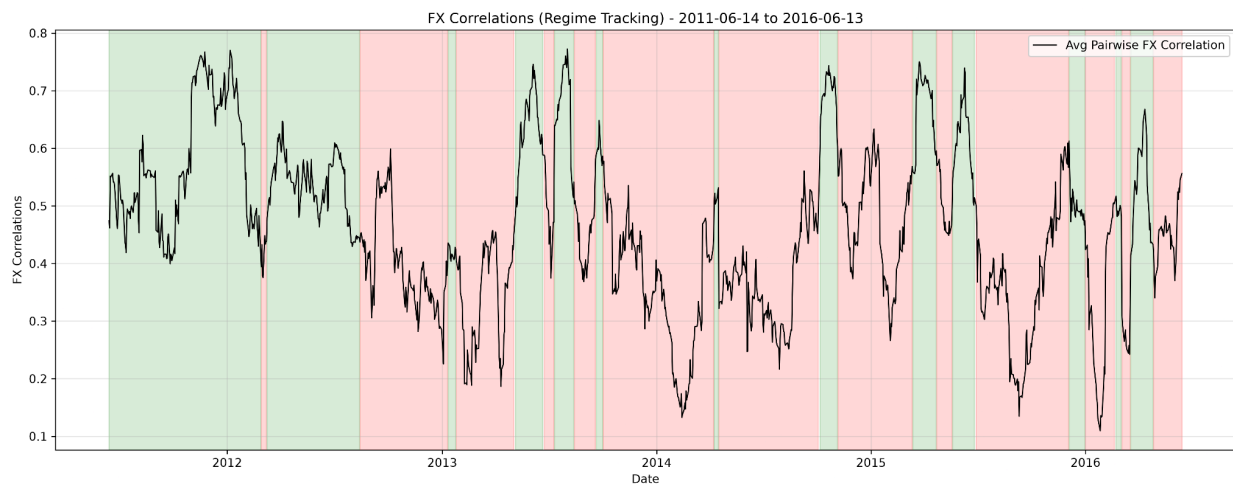
Each observation is assigned to one of two latent states inferred directly from market behavior.

Results

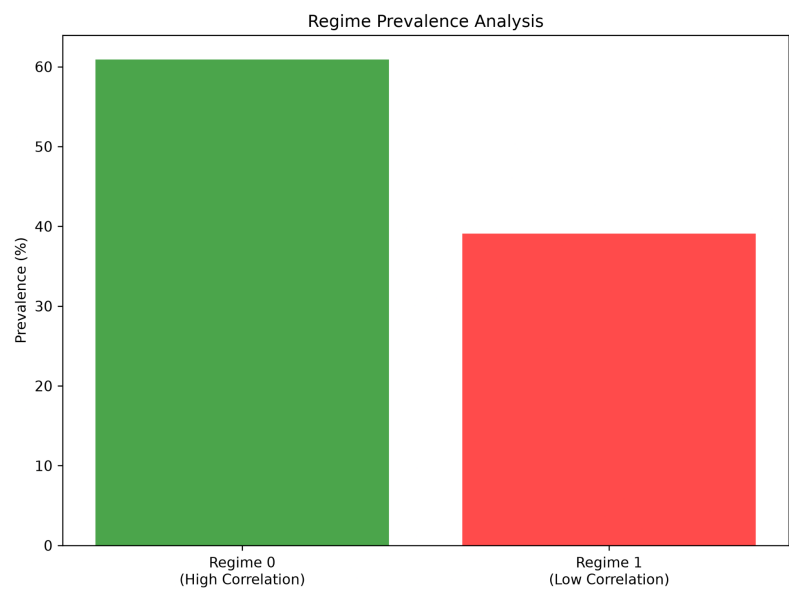
Average FX Correlations Chart

Covering a full sample from 2006 to 2026 with colored labels indicating high FX correlation (green) and low FX correlation (red)

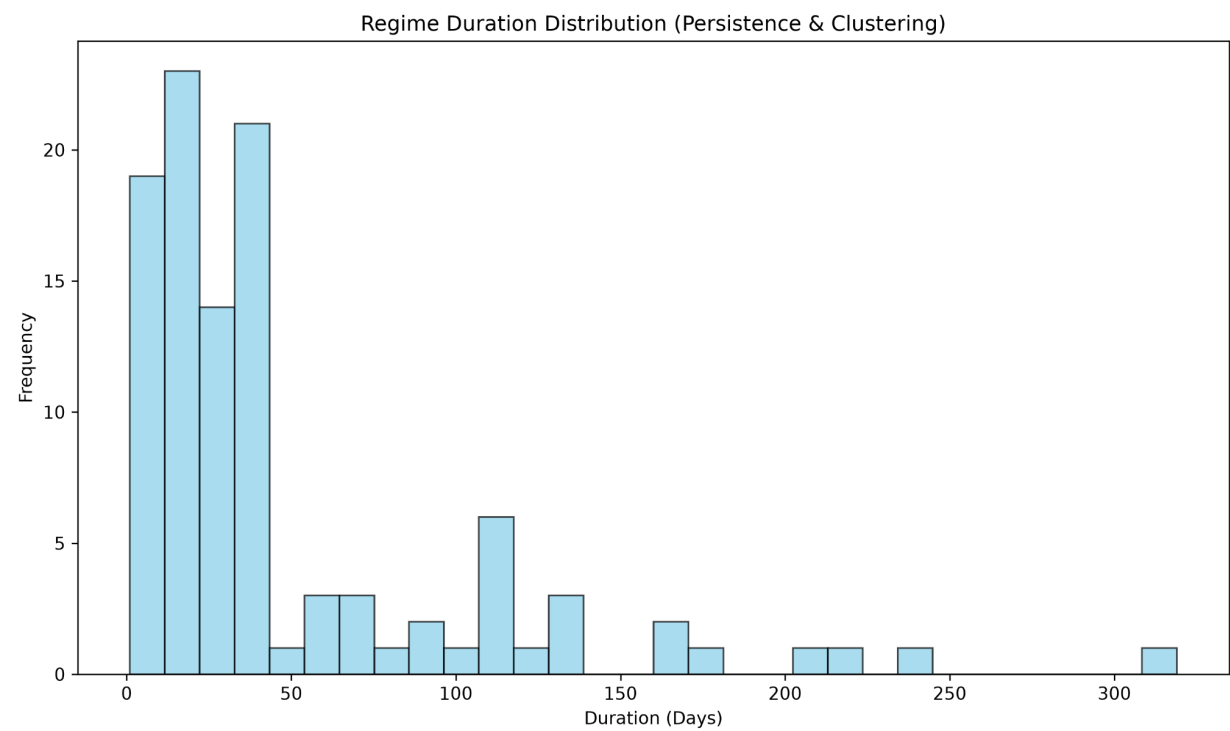




Regime Prevalence Bar Chart



Regime Duration Distribution Histogram



Interpretation & Discussion

The model identified two highly persistent market states.

Persistence

Transition Matrix

From	To Regime 0	To Regime 1
Regime 0	98.33%	1.67%
Regime 1	2.59%	97.41%

This indicates that regime transitions occur infrequently.

Average regime durations exceeded 49 trading days, suggesting that these states represent meaningful structural environments rather than short-term fluctuations.

Correlation Structure

Regime 1 exhibited the highest average market correlation:

0.5798

versus

0.3937 for Regime 0.

This suggests materially different market dynamics between states.

Higher correlation environments imply:

- Greater market synchronization
- Reduced diversification benefits
- Stronger systemic participation

Lower correlation environments imply:

- Greater dispersion
- Increased relative-value opportunities
- More independent currency behavior

Current Regime

Latest observation:

June 13, 2026

Current State:

Regime 1

This indicates the market remains in the higher-correlation environment identified by the model.

Limitations

Correlations Only

The model uses market behavior exclusively and does not incorporate:

- Interest rates
- Inflation
- GDP
- Employment
- Central bank policy

Two-State Assumption

Financial markets likely contain more than two structural environments.

Future research should evaluate:

- 3-state models
- 4-state models
- Hierarchical regime structures

No Forward Return Testing

This study identifies regimes but does not yet test:

- Future returns
- Volatility forecasts
- Strategy performance by regime

Author

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Background

- Former Research Analyst – Institutional Shareholder Services
- Former Senior Research Analyst (M&A) – FactSet
- Equity Investor Since 2015
- Futures/Forex/Crypto Trader Since 2020
- Quantitative Market Researcher

Related Research

- FX Correlation Regime Predictability Analysis
- Global Macro Regime Analysis
- NQ/ES Liquidity Sweep Studies
- Session Behavior Analytics
- SMT Divergence Research
- Trend Continuation Probability Analysis

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